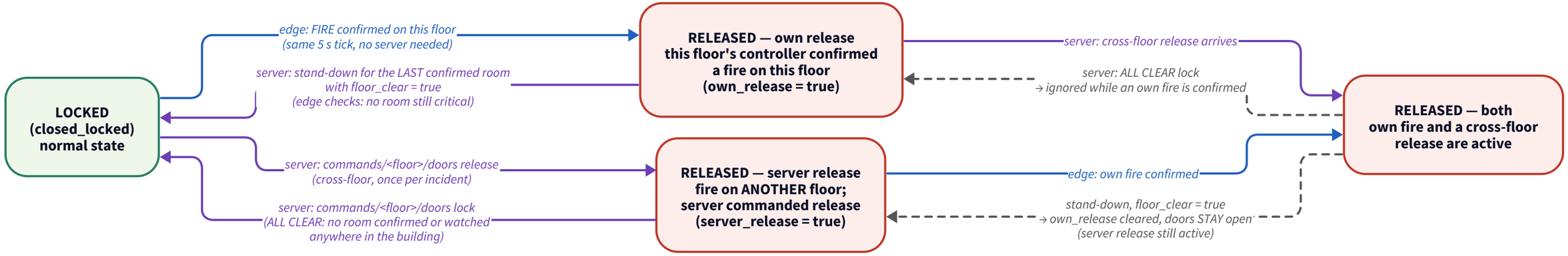
Fire doors of ONE floor — who may open them, who may close them



Rules that make this survive restarts and failover

- Opening is cheap and safe: either tier may release; the edge never waits for the server to open its own floor.
- Closing needs agreement: the edge re-locks only when the *floor* is clear, and refuses a stand-down while a room still reads critical (smoke ≥ 10 and CO ≥ 20) — `stand_down_refused` is sent back and the room is re-confirmed.
- A controller that just started (restart, failover) has no memory of what it released. Every stand-down therefore carries `floor_clear` and `relock`, computed by the server from its own view; `relock = true` makes a fresh controller lock even though it never opened.
- The ALL CLEAR `lock` is unconditional for a floor without an own fire, idempotent, and skipped by a controller that already wrote that state in this process.
- A floor under server takeover is actuated directly through BuildSim; the same state machine applies, with the server playing both roles.